

On the Geometry of On-Policy Distillation

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ABSTRACT

On-policy distillation (OPD) is increasingly used to improve large language model reasoning, but its training dynamics remain poorly understood. We characterize the trajectory of OPD updates in parameter space and compare it with supervised fine-tuning (SFT) and reinforcement learning with verifiable rewards (RLVR). A suite of parameter-space diagnostics consistently places OPD in a relaxed off-principal regime: compared with SFT, its updates affect fewer weights and avoid principal directions more strongly, while compared with RLVR, they remain less tightly constrained. Beyond this static localization, OPD exhibits subspace locking: its cumulative updates rapidly enter a narrow low-dimensional channel. Constraining training to the update subspace formed early in training preserves OPD performance but substantially degrades SFT, indicating that the locked subspace is functionally sufficient for OPD. Control experiments further show that sparsifying the update tokens and shifting rollout generation off-policy preserve the rank dynamics, whereas mixing the OPD objective with RLVR changes them. Overall, these results suggest that OPD is not merely an intermediate point between SFT and RLVR, but induces its own update geometry in parameter space.

1 Introduction

Large reasoning models (LRMs) have substantially advanced complex mathematical and programming reasoning in large language models [5, 15, 17]. Post-training is a central driver of this progress. Beyond supervised fine-tuning (SFT) [20] on offline demonstrations and reinforcement learning with verifiable rewards (RLVR) [5, 17, 23] from sparse outcome signals, on-policy distillation (OPD) has recently emerged as a complementary paradigm: it trains a student on its own sampled trajectories under dense token-level guidance from a stronger teacher [1, 11].

Despite its empirical utility, the parameter-space dynamics of OPD remain poorly understood. Prior analyses show that SFT and RLVR leave distinct geometric footprints: SFT induces dense, principal-aligned updates [10], whereas RLVR produces sparse, off-principal updates that better preserve pretrained spectral structure [13, 24]. OPD combines features of both: dense token-level distillation resembles SFT, while on-policy sampling and policy-gradient optimization connect it to RLVR [1]. This makes its parameter tra-

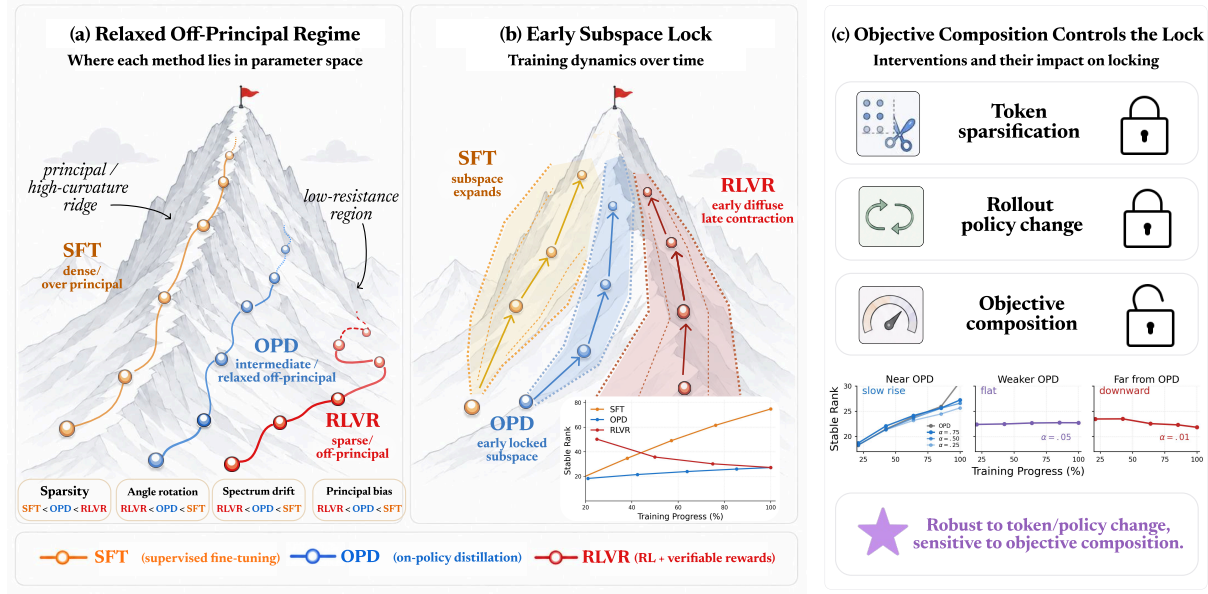


FIGURE 1 Optimization geometry of OPD compared with SFT and RLVR. (a) OPD occupies a relaxed off-principal regime between dense principal-aligned SFT updates and sparse off-principal RLVR updates. (b) OPD rapidly enters a locked low-dimensional subspace during training. (c) The lock is robust to token and rollout perturbations, but sensitive to objective composition.

jectory difficult to infer from either paradigm alone. Thus, we study OPD through three research questions:

RQ1 *Where does OPD lie within the parameter-space spectrum between SFT and RLVR?*

RQ2 *What intrinsic update trajectory does OPD follow during training?*

RQ3 *Which component of OPD controls this trajectory?*

OPD occupies a relaxed off-principal regime (§3). Answering RQ1, we locate OPD relative to SFT and RLVR using a suite of parameter-space diagnostics. The results show that across update sparsity, spectral drift, principal-subspace rotation, and update localization, OPD consistently falls between SFT and RLVR with a bias toward the RLVR side. We refer to this region as a *relaxed off-principal regime*: OPD is more selective and geometry-preserving than SFT, yet less constrained than RLVR. We further interpret this positioning through a relaxed Three-Gate view [24]: OPD retains RLVR’s geometry-preserving update bias, but dense teacher supervision broadens the set of active directions and makes more coordinates visibly update.

OPD learns through subspace locking (§4). Answering RQ2, we move beyond endpoint localization and characterize how OPD updates evolve during training: we track cumulative updates ΔW_t across checkpoints using effective dimension, update scale, and spectral shape diagnostics. This trajectory-level analysis shows that OPD rapidly enters a narrow low-dimensional update band, while SFT expands and RLVR contracts. This is not a vanishing-update artifact: OPD has a substantially larger cumulative update norm than

RLVR while ending with comparable stable rank. We then test whether this low-dimensional channel is stable and functional: subspace similarity shows early alignment with the final update channel, and constraining subsequent training to this early subspace preserves OPD performance while degrading SFT. These results identify OPD subspace locking: an early-emerging, persistent, and functionally sufficient low-dimensional update channel.

Objective composition controls subspace locking (§5). Answering RQ3, we test which part of OPD maintains the locked trajectory by perturbing three factors that distinguish OPD from standard SFT or RLVR: token supervision density, rollout policy, and objective composition. Token sparsification and off-policy rollouts preserve the OPD rank trajectory, changing update scale at most. In contrast, objective-level interpolation exposes a boundary of the locked regime, where the rank dynamics depart from the trajectory. These controls indicate that the lock is robust to runtime perturbations but sensitive to objective composition.

Together, our findings provide a parameter-space account of OPD post-training. OPD is not merely an endpoint interpolation between SFT and RLVR; it follows a distinct update trajectory characterized by relaxed off-principal localization, early subspace locking, and sensitivity to objective composition. We discuss how these diagnostics can guide future geometry-aware OPD algorithms in Section 6.

2 Related Work

Post-training for large language models. Post-training large language models for reasoning has largely followed three paradigms. Supervised fine-tuning (SFT) trains on offline demonstrations with cross-entropy objectives [3, 7, 20]. Reinforcement learning methods, including RLHF and reinforcement learning with verifiable rewards (RLVR), optimize policies using preference or outcome-level feedback [5, 16, 17, 25].

More recently, on-policy distillation (OPD) has emerged as a complementary route: it trains on student-generated rollouts while using a stronger teacher to provide dense token-level guidance [11, 19]. Recent work studies OPD from the perspectives of empirical failure modes, scaling recipes, and multi-teacher distillation [2, 4, 9, 22]. These accounts characterize when OPD works behaviorally, but leave its parameter-space dynamics largely unexamined.

Post-training weight geometry. Recent studies reveal a fundamental optimization dichotomy: SFT induces dense weight updates that distort the pretrained spectral structure along principal directions [10], whereas online RL targets highly localized, off-principal subnetworks [13]. This RL bias can be viewed as a conservative projection that limits policy-level KL drift [18, 21], with pretrained model geometry steering the trajectory toward low-curvature, spectrum-preserving subspaces [24].

These studies establish the SFT–RLVR geometric split, but leave open where OPD lies within this spectrum and how its updates evolve during training. We study this missing regime.

Base Model	Finetuned (FT) Model	Algorithm	Data	sparsity _{bf16}
<i>Controlled comparison: SFT < OPD < RLVR</i>				
Qwen3-8B-Base	Qwen3-8B-SFT	SFT	Math	8.1%
Qwen3-8B-SFT	OPD-8B-T32B	OPD	Math	51.6%
Qwen3-8B-SFT	RLVR-8B	GRPO	Math	77.2%
<i>OPD robustness across teacher / student / data</i>				
Qwen3-4B-SFT	OPD-4B-T8B	OPD	Math	50.3%
Qwen3-4B-SFT	OPD-4B-T14B	OPD	Math	51.1%
Qwen3-4B-SFT	OPD-4B-T32B	OPD	Math	51.7%
Qwen3-14B-SFT	OPD-14B-T32B	OPD	Math	56.6%
Qwen3-8B-SFT	OPD-8B-T32B-Code	OPD	Code	57.1%
Qwen3-8B-SFT	OPD-8B-T32B-MoE	OPD	Math	48.6%
<i>Published reference points</i>				
Qwen3-8B-Base	Klear-Reasoner-8B-SFT	SFT	Math-Code	0.6%
Qwen3-14B-Base	UniReason-14B-think-SFT	SFT	Mixed [†]	18.8%
Klear-Reasoner-8B-SFT	Klear-Reasoner-8B	GRPO	Math-Code	69.9%
Qwen3-8B-Base	GT-Qwen3-8B-Base	GRPO	Math	79.0%
Qwen3-4B	Polaris-4B-Preview	DAPO	Math	79.9%

TABLE 1 bf16-aware update sparsity. OPD lies between SFT and RLVR in the controlled Qwen3-8B comparison, and remains stable across OPD variants. Published checkpoints are used only as external reference points. In OPD variant names, TxB denotes a teacher model with xB parameters, e.g., OPD-4B-T8B uses a 4B student and an 8B teacher. Mixed[†] denotes a mixture of math, code, STEM, logic, and instruction data.

3 Locating OPD in Parameter Space

We first ask whether on-policy distillation (OPD) should be understood as an SFT-like distillation method, an RLVR-like online optimizer, or a distinct parameter-space regime. To answer this, we locate OPD within the SFT-RLVR spectrum using diagnostics over update support, subspace rotation, spectral drift, and update localization. This section first defines these diagnostics, then uses them to compare OPD with SFT and RLVR under a controlled setting, and finally interprets the observed positioning through a relaxed Three-Gate view.

Experimental setup. We analyze Qwen3-family checkpoints [22] spanning teacher-student gaps, data domains, and dense/MoE teachers. Our main comparison centers on Qwen3-8B: OPD and RLVR share the same SFT anchor and math-domain prompt distribution, while SFT is measured as the preceding supervised adaptation stage from the pretrained base. Full experimental details are provided in Appendix C.

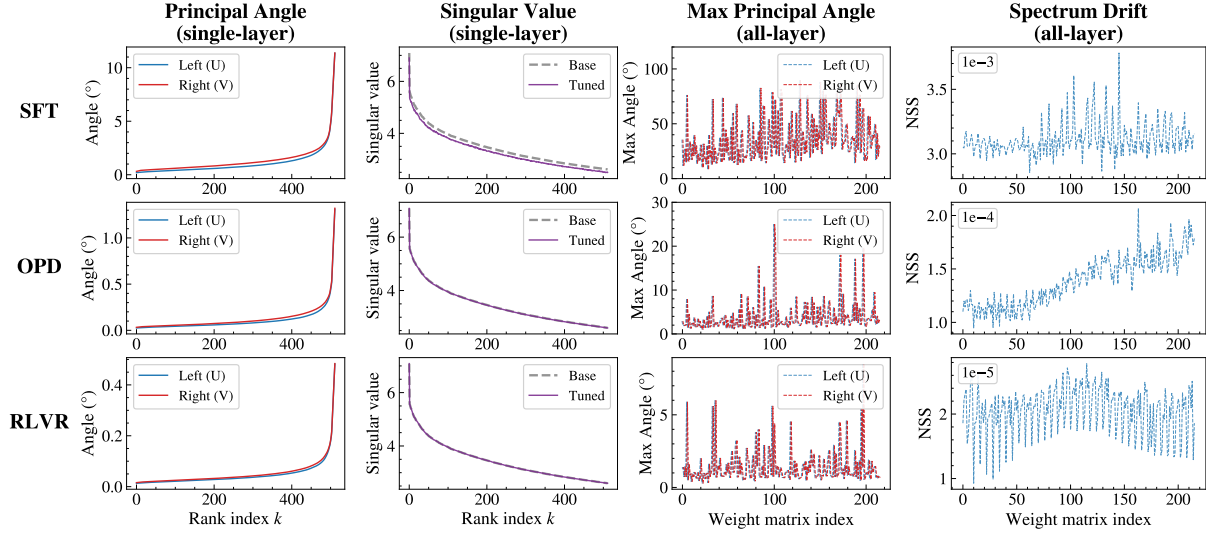


FIGURE 2 Parameter-space diagnostics. SFT induces larger subspace rotation and spectral drift, RLVR preserves the pretrained geometry most strongly, and OPD lies between them. Here, k denotes the rank index of principal angles or singular values; the all-layer panels enumerate analyzed weight matrices across layers and module types.

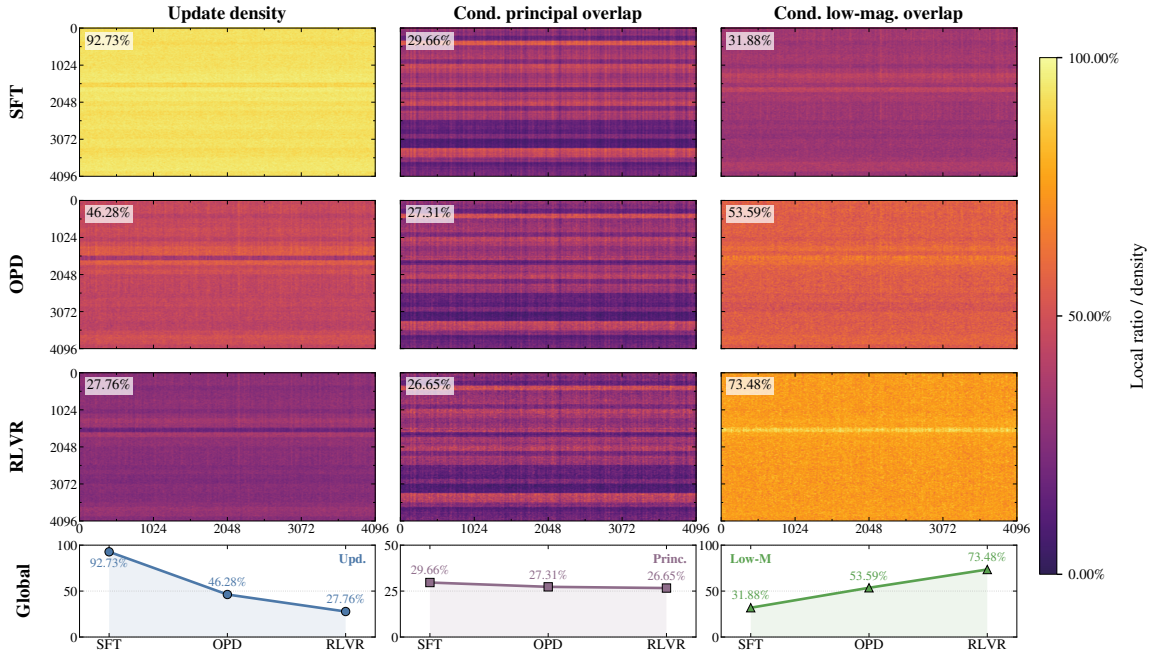


FIGURE 3 Update-mask localization. We compare where bf16-visible updates land relative to principal and low-magnitude masks. OPD shifts updates away from principal weights and toward low-magnitude regions, while remaining less selective than RLVR.

5.1 Parameter-Space Diagnostics

Following recent work on post-training weight geometry [24], we use four diagnostics to measure where updates occur, how far they rotate pretrained subspaces, how much they

distort the spectrum, and whether they land on principal or low-magnitude weights. Let W_0 denote the initialization, W_+ the post-training weight matrix, and $\Delta W = W_+ - W_0$.

Update sparsity. Under bfloat16 precision, small updates can be absorbed by rounding. We treat scalar weights $w_i, \hat{w}_i \in \mathbb{R}$ as unchanged if

$$|\hat{w}_i - w_i| \leq \eta \max(|w_i|, |\hat{w}_i|), \quad \eta = 10^{-3}.$$

The bf16-aware update sparsity is

$$\text{sp}(W_0, W_+) = 1 - \frac{1}{n} \sum_{i,j} \mathbb{1}[W_{+,ij} \not\approx_{\eta} W_{0,ij}], \quad (1)$$

where n is the number of entries. Larger values indicate fewer visible weight changes.

Principal-angle rotation. We measure rotation of the dominant singular subspaces by the principal angles between the top- k subspaces of W_0 and W_+ :

$$\begin{aligned} \cos \theta_i(U) &= \sigma(U_{0,k}^\top U_{+,k}), \\ \cos \theta_i(V) &= \sigma(V_{0,k}^\top V_{+,k}), \end{aligned} \quad (2)$$

where $U_{\cdot,k}$ and $V_{\cdot,k}$ denote the top- k left and right singular vectors. Smaller angles indicate stronger preservation of the pretrained dominant subspaces.

Spectral drift. We quantify distortion of the singular-value spectrum using the normalized spectral shift:

$$\text{NSS}(W_0, W_+) = \frac{\|\sigma(W_+) - \sigma(W_0)\|_2}{\|\sigma(W_0)\|_2}, \quad (3)$$

where $\sigma(\cdot)$ denotes singular values in descending order. Smaller NSS indicates stronger spectral preservation.

Update-mask overlap. To localize visible updates, we compare the update mask

$$M = \{(i, j) : \text{bf16}(W_+)_{ij} \neq \text{bf16}(W_0)_{ij}\}$$

with two masks derived from W_0 . The principal mask M_{princ} contains the top- α fraction of entries in the rank- k SVD reconstruction of W_0 , serving as a proxy for high-curvature regions [10]. The low-magnitude mask M_{low} contains the bottom- α fraction of entries by $|W_0|$. For $M_\star \in \{M_{\text{princ}}, M_{\text{low}}\}$, we report

$$\text{Overlap}(M_\star, M) = \frac{|M_\star \cap M|}{|M|},$$

with random baseline α . Sub-random overlap with M_{princ} indicates depletion from principal weights, while super-random overlap with M_{low} indicates concentration in low-magnitude regions.

3.2 OPD Occupies a Relaxed Off-Principal Regime

Taken together, these diagnostics place OPD in an intermediate but RLVR-biased region of the parameter-space spectrum. OPD is more selective and geometry-preserving than SFT, yet less constrained than RLVR. We call this regime *relaxed off-principal*.

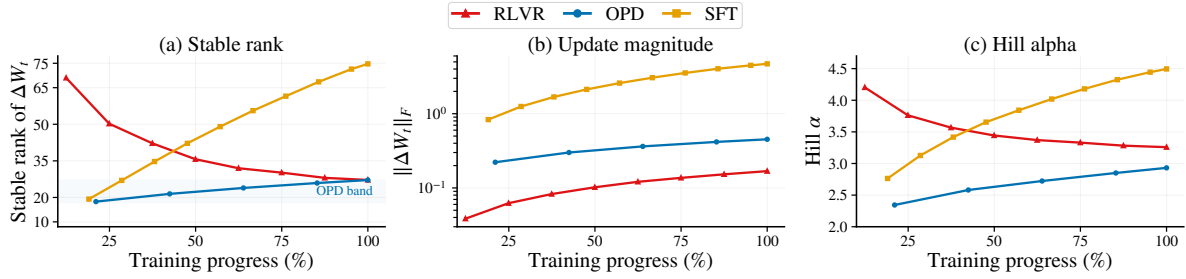


FIGURE 4 Intrinsic update geometry. We track cumulative updates ΔW_i . OPD stays in a narrow stable-rank band, whereas SFT expands and RLVR contracts. Frobenius norms rule out a small-update explanation: OPD moves more than RLVR while ending with comparable stable rank. Hill estimates provide an auxiliary spectral-shape check.

Update sparsity. In the controlled Qwen3-8B comparison, SFT leaves only 8.1% of weights unchanged at bf16 precision, while RLVR leaves 77.2% unchanged. OPD lies between them at 51.6%. This pattern is stable across OPD variants: teacher scale, student scale, code data, and a MoE teacher keep OPD within 48.6%–57.1% sparsity. Published reference checkpoints exhibit the same SFT–RLVR separation (Table 1).

Subspace rotation and spectral drift. SFT induces the largest rotation of the pretrained singular subspaces: in the representative layer, its principal angles rise above 10° . OPD shows smaller but nonzero rotation, with representative-layer angles around 1° , while RLVR exhibits the smallest rotations below 0.5° . Spectral drift follows the same ordering: SFT is at the 10^{-3} level, OPD at the 10^{-4} level, and RLVR at the 10^{-5} level (Figure 2).

Update localization. Conditioned on bf16-visible updates, global update density decreases from SFT to OPD to RLVR: 92.73%, 46.28%, and 27.76%. Principal-mask overlap also decreases monotonically, placing OPD and RLVR below the 30% random baseline. Low-magnitude overlap shows a stronger separation: 31.88% for SFT, 53.59% for OPD, and 73.48% for RLVR (Figure 3).

Across all three diagnostics, OPD follows the same off-principal direction as RLVR, but with weaker selectivity and larger visible support.

3.5 A Relaxed Three-Gate Account of OPD

We next explain why the empirical regime above is both off-principal and relaxed by extending the Three-Gate account of RLVR optimization [24]. In this view, RLVR updates become off-principal through three filters: (1) a distributional anchor that limits update size, (2) pretrained model geometry that routes bounded updates away from dominant spectral directions, and (3) finite-precision realization that determines which coordinates become visibly changed. We argue that OPD preserves this gated structure, but relaxes it through dense token-level teacher supervision. Full objective forms and derivation details are provided in Appendix D.

Signal granularity. The key difference is the granularity of the training signal. Let

$$\phi_t = \nabla_{\theta} \log p_{\theta}(y_t | x, y_{<t})$$

denote the token score. A broad class of post-training updates can be written as

$$g = \sum_{t=1}^T a_t \phi_t. \quad (4)$$

The paradigms differ in the coefficients a_t . In RLVR, the reward signal is sequence-level, so $a_t = A(y)$ is shared across tokens. In OPD, a_t varies across tokens according to local teacher–student disagreement. Thus, OPD retains the on-policy form of RLVR, but replaces a scalar credit signal with dense token-level supervision.

Gate I: distributional anchor. OPD remains anchored. Under a local quadratic budget, the update norm is bounded as

$$\|\Delta W\|_F \leq \sqrt{\frac{2\delta_W}{\mu_W}}. \quad (5)$$

The difference is how this budget is used. RLVR spends it through a sequence-level reward signal, whereas OPD uses teacher-token distributions to provide denser descent directions within the anchored region. This yields larger effective updates than RLVR while remaining far below SFT-style unconstrained rewriting.

Gate II: model geometry. A bounded update is still shaped by pretrained model geometry. OPD therefore does not freely rewrite the dominant spectral structure as SFT does. The relaxation appears in the update covariance:

$$\mathbb{E}[gg^T] = \sum_{t,t'} \mathbb{E}[a_t a_{t'} \phi_t \phi_{t'}^T]. \quad (6)$$

RLVR couples all token scores through a shared scalar coefficient, whereas OPD uses heterogeneous token coefficients. This broadens the accessible directional support while keeping the update geometry-steered, matching the observed pattern that OPD preserves spectral structure more than SFT but less strictly than RLVR.

Gate III: precision realization. The bf16 realization gate is unchanged. A coordinate becomes visibly updated only when

$$M_{ij} = 1 \left[|\Delta W_{ij}| \gtrsim \frac{1}{2} \text{ULP}_{\text{bf16}}(W_{0,ij}) \right]. \quad (7)$$

Larger geometry-constrained updates allow more OPD coordinates to pass the bf16 realization threshold, lowering sparsity while preserving the off-principal bias.

Implication. OPD occupies a relaxed off-principal regime: geometry-steered like RLVR, but less selective under dense teacher supervision. This suggests that OPD recipes should regulate update geometry, rather than treating token-level supervision density as the primary design axis.

4 Subspace Locking in OPD

Section 3 located on-policy distillation (OPD) within the SFT-RLVR parameter-space spectrum. We now ask whether this positioning is only an endpoint property, or whether OPD follows a distinct update trajectory during training. We study the cumulative update $\Delta W_t = W_t - W_0$ across checkpoints and show that OPD rapidly enters a persistent low-dimensional update channel.

4.1 A Low-Dimensional Update Band

We first characterize the effective dimension of ΔW_t . For each matrix, we compute the stable rank

$$\text{srnk}(\Delta W_t) = \frac{\|\Delta W_t\|_F^2}{\|\Delta W_t\|_{\text{op}}^2}, \quad (8)$$

and average over all analyzed weight matrices. This measures how many dominant singular directions carry the update energy.

Figure 4(a) shows that OPD stays within a narrow low-rank band throughout training. SFT progressively expands its update subspace, while RLVR contracts toward a low-dimensional endpoint. Thus, OPD is not a temporal interpolation between the two: it enters the low-dimensional regime early and remains there.

To rule out a trivial small-update explanation, we complement stable rank with update-scale and spectral-shape diagnostics. Frobenius norms in Figure 4(b) show that OPD accumulates a substantially larger update than RLVR while ending with comparable stable rank. Hill tail estimates in Figure 4(c) provide an auxiliary check on the singular-value profile: OPD evolves mildly, whereas SFT increases sharply and RLVR decreases over training. Together, these diagnostics show that OPD’s low-dimensionality is not a byproduct of small parameter movement, but a bounded spectral profile of a nontrivial update. Full metric definitions are given in Appendix E.

4.2 Stability and Functional Sufficiency

Early subspace emergence. Low dimensionality alone does not imply subspace locking: a trajectory may remain low rank while rotating through different low-dimensional subspaces. We therefore test when the final low-dimensional update channel emerges. Let $V_K(t)$ denote the top- K right singular subspace of ΔW_t . We compare each checkpoint to the final subspace $V_K(t_{\text{end}})$:

$$\text{Sim}_K(t, t_{\text{end}}) = \frac{1}{K} \left\| V_K(t)^\top V_K(t_{\text{end}}) \right\|_F^2. \quad (9)$$

Values near 1 indicate that the update already uses the singular directions of the final update channel.

Figure 5 shows that OPD aligns with its final V_{16} subspace from the first measured checkpoint, whereas SFT and RLVR converge more gradually. Thus, OPD’s low-dimensional channel emerges early rather than being assembled late.

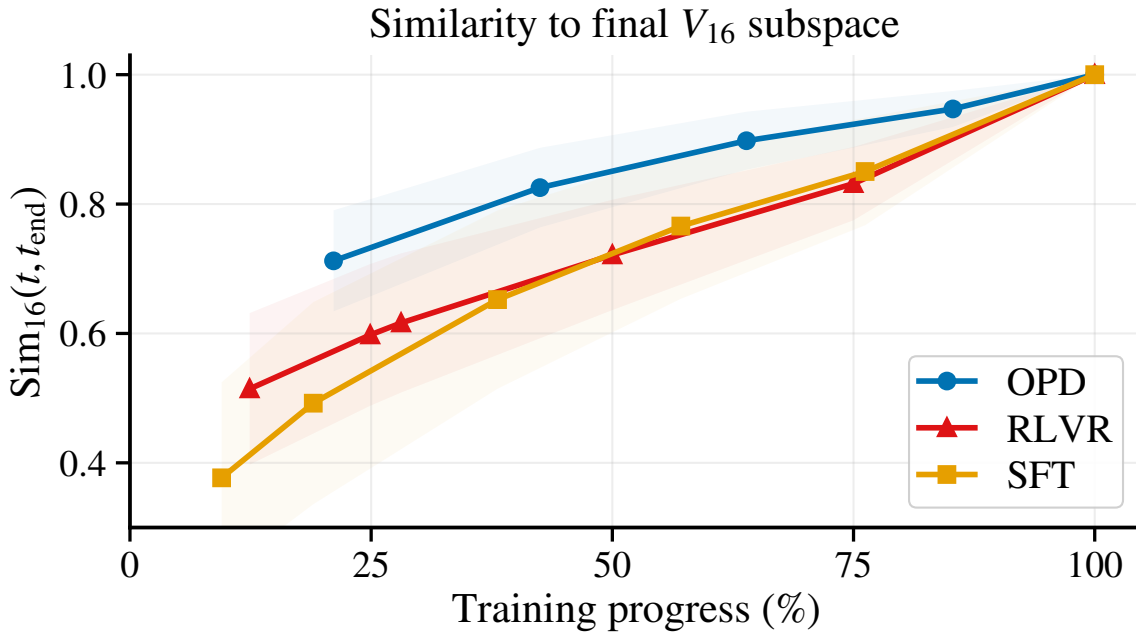


FIGURE 5 Subspace emergence. Top-16 subspace similarity to the final update shows that OPD locks onto its final update channel earlier than SFT and RLVR.

Functional sufficiency. We next ask whether this channel is only descriptive or also sufficient for learning. Motivated by the early stable-rank scale of OPD, we use $K = 16$ as a stringent bottleneck: it is close to, but slightly below, the effective dimension observed near the projection point. We then constrain each gradient matrix g to the early right singular subspace:

$$g \leftarrow gV_{16}V_{16}^T. \quad (10)$$

For both OPD and SFT, V_{16} is extracted from the top right singular vectors of ΔW_t around 20% training progress, and training then resumes under this rank-16 constraint.

Figure 6 shows that OPD is essentially unchanged under the rank-16 bottleneck, indicating that the early low-dimensional channel is sufficient for OPD training. The same constraint degrades SFT over the matched window, confirming that this sufficiency is not a generic property of the projection dimension. The same qualitative pattern holds across additional reasoning benchmarks (Appendix F, Figure 8).

Implication. OPD subspace locking identifies an early-emerging update channel that is both persistent and sufficient for training. Future OPD implementations should therefore monitor and exploit this channel, rather than treating low-dimensionality as a post-hoc spectral statistic only.

5 What Controls Subspace Locking?

Section 4 shows that on-policy distillation (OPD) learns through an early-emerging update channel sufficient for training. We now ask what maintains this channel by perturbing three candidate factors: token-level teacher supervision, rollout policy, and objective composition. We use stable rank as the primary diagnostic, with details in Appendix G.

5.1 Runtime Perturbations Preserve the Lock

We first perturb runtime sources while keeping the objective fixed. Token sparsification retains either top-KL or randomly selected tokens at 25% and 50% density. Figure 7(a) shows that all sparsified variants closely track the OPD stable-rank trajectory. Even random 25% retention changes update scale more than spectral shape, indicating that the low-rank profile is not localized to a small set of high-KL tokens.

Off-policy rollout perturbations also leave the stable-rank trajectory nearly unchanged (Figure 7(b)). Although off-policy OPD produces a modestly larger update norm, its stable rank remains matched to on-policy OPD, showing robustness to rollout-policy changes.

5.2 Objective Mixing Changes the Rank Dynamics

We perturb objective composition by interpolating the OPD and RLVR advantage signals in our GRPO-style policy-gradient implementation:

$$A_i^{(\alpha)} = \alpha A_{i,\text{OPD}} + (1 - \alpha) A_{i,\text{RLVR}}. \quad (11)$$

Unlike token sparsification or off-policy generation, this changes the gradient source rather than only the sampled data.

Figure 7(c) shows a clear split. OPD-dominant mixtures retain the OPD-like stable-rank trajectory. When the OPD component becomes weak, the trajectory no longer follows the baseline and enters a distinct spectral regime. Objective composition therefore changes the rank dynamics in a way that runtime perturbations do not.

Mechanistic view. The controls separate sample perturbations from objective perturbations. OPD gradients can be written as token-level sums,

$$g_{\text{OPD}} = \sum_t J_t^\top \delta_t, \quad \tilde{g}_{\text{OPD}} = \sum_t m_t J_t^\top \delta_t, \quad (12)$$

where J_t is the local Jacobian, δ_t is the teacher–student token discrepancy, and m_t is a token mask. If token gradients share a dominant update subspace, masking primarily rescales the second moment,

$$\mathbb{E}[\tilde{g}\tilde{g}^\top] \approx c \mathbb{E}[g_{\text{OPD}}g_{\text{OPD}}^\top] + \text{noise}, \quad (13)$$

so the leading spectral directions are preserved. Off-policy rollouts change the sampling distribution, but retain the same teacher-token gradient source. Objective mixing changes

the source itself:

$$\begin{aligned} g_\alpha &= \alpha g_{\text{OPD}} + (1 - \alpha) g_{\text{RLVR}}, \\ \Sigma_\alpha &\approx \alpha^2 \Sigma_{\text{OPD}} + (1 - \alpha)^2 \Sigma_{\text{RLVR}} \\ &\quad + \alpha(1 - \alpha) \Sigma_{\text{cross}}. \end{aligned} \tag{14}$$

Thus, weakening the OPD term can change the dominant covariance geometry, explaining why objective mixing, unlike token thinning or rollout changes, breaks the OPD-like rank trajectory.

Implication. Subspace locking is objective-sensitive rather than runtime-induced. Effective OPD recipes should regulate objective-induced update geometry, not only token coverage or rollout generation.

6 Discussion & Conclusion

Our analysis identifies on-policy distillation (OPD) as a distinct parameter-space regime rather than a simple endpoint interpolation between SFT and RLVR. OPD occupies a relaxed off-principal region, but its training trajectory further exhibits subspace locking: cumulative updates rapidly enter a small, persistent low-dimensional channel that is sufficient to preserve training progress.

These findings suggest a key guiding principle for future OPD algorithms: design OPD as *geometry control*, not merely as denser token supervision. Effective recipes should monitor the locked update channel and use objective composition as the primary lever when this geometry drifts, with token selection, rollout policy, and teacher scale tuned through their effect on update geometry. Such geometry-aware control may make OPD more stable, interpretable, and transferable by preserving the update channel that supports learning while avoiding unnecessary parameter-space drift.

Limitations

Our analysis focuses on controlled Qwen3-family reasoning settings. While this design isolates the effect of training paradigms, the observed geometry may vary across other model families, modalities, and task distributions.

Our diagnostics characterize parameter-space trajectories from stored checkpoints. The Three-Gate account and covariance analysis should therefore be viewed as mechanistic explanations consistent with the evidence, rather than complete causal or formal theories of on-policy distillation (OPD) optimization.

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A AI Usage

We used ChatGPT for writing assistance, including language polishing, LaTeX formatting, organization suggestions, and refinement of presentation. We also used it to assist with drafting plotting and analysis scripts. All scientific claims, experimental designs, code, figures, numerical results, and final manuscript content were reviewed, verified, and edited by the authors. No generative AI system was used as an author, to produce experimental results, or to make autonomous scientific decisions.

B Artifact Use

We use publicly available model checkpoints, datasets, and benchmarks only for research evaluation and analysis. We cite the corresponding creators in the main text or appendix and use these artifacts consistently with their intended research use. We checked the public license or usage terms of the major artifacts where available, including Qwen3, DAPO-Math-17k, Dolci-Think SFT, DeepCoder, and LiveCodeBench. We do not redistribute the original artifacts.

C Experimental Details

C.1 Training Setup

This section summarizes the training setups used in the main parameter-space analyses. Table 2 gives the shared setup. Tables 3, 4, and 5 report the default supervised finetuning (SFT), on-policy distillation (OPD), and reinforcement learning with verifiable rewards (RLVR) settings, respectively. OPD and RLVR are initialized from the same SFT anchor, while the SFT trajectory is measured from the pretrained base checkpoint. We use Qwen3 checkpoints [22]; math-domain OPD/RLVR runs use dap0-math-17k [23], and the SFT anchor is trained on Dolci-Think SFT data [14].

Item	Shared setting
Model family	Qwen3
Main backbone	Qwen3-8B
Evaluation	AIME 2024 avg@8
Precision	bf16 training
Gradient accumulation	fp32
Attention softmax	fp32
Hardware	H200 80GB GPUs
Dropout	0.0

TABLE 2 Shared settings used in the main training runs.

Endpoint selection. We choose the main analyzed endpoints according to the evaluation curves observed during training. The SFT, OPD, and RLVR runs showed little additional improvement near the selected checkpoints, corresponding approximately to 5k, 300, and 1k training steps, respectively. We use these rounded horizons as the main comparison points for interpretability and reproducibility, rather than tuning endpoints to optimize the parameter-space diagnostics.

Geometry origins. For OPD and RLVR, W_0 is the shared SFT anchor `iter_0005375`. For SFT, W_0 is the pretrained Qwen3-8B-Base checkpoint. All updates are computed as $\Delta W = W_+ - W_0$

Item	SFT default setting
Initialization	Qwen3-8B-Base
Objective	Supervised next-token prediction
Loss mask	Qwen3 loss mask
Training data	Dolci-Think SFT data
Epochs	5
Final checkpoint	iter_0005375
Global batch size	256
Learning rate	10^{-5}
Schedule	Cosine decay to 10^{-6}
Warmup	10% of training steps
Optimizer	Adam
Weight decay	0.1
Adam betas	(0.9, 0.95)
Save interval	256 steps
Eval interval	256 steps
Role	Produces the shared SFT anchor

TABLE 3 Default SFT setting used to produce the shared anchor checkpoint.

relative to the corresponding stage initialization. Thus, OPD and RLVR share a common origin in the parameter-space comparison.

C.2 OPD Variant Setup

All OPD variant runs follow Table 4 unless the changed factor is listed explicitly. Table 6 summarizes the student, teacher, and changed factor for each variant. Additional variant-specific notes are provided in Table 7.

The code-domain variant uses DeepCoder data [12]; evaluation is conducted on Live-CodeBench v5 [8].

C.5 Parameter-Space Diagnostic Implementation

All diagnostics are computed offline on saved checkpoints. Each analysis loads a pair of checkpoints (W_0, W_+) and computes $\Delta W = W_+ - W_0$. For OPD and RLVR, W_0 is the shared SFT anchor; for SFT, W_0 is the pretrained Qwen3-8B-Base model. Metrics are computed per matrix and then averaged across matrices unless otherwise specified.

bf16-aware update sparsity. Both W_0 and W_+ are cast to bf16 and then back to fp32 before comparison. A scalar entry is considered unchanged when

$$|W_{+,ij}^{\text{bf16}} - W_{0,ij}^{\text{bf16}}| \leq \eta \max(|W_{0,ij}^{\text{bf16}}|, |W_{+,ij}^{\text{bf16}}|),$$

$$\eta = 10^{-3}.$$

Item	OPD default setting
Initialization	SFT anchor iter_0005375
Student	Qwen3-8B
Teacher	Qwen3-32B
Training data	dapo-math-17k
Rollout policy	Student-generated, on-policy
Rollout prompts	300 per step
Samples per prompt	8
Rollout temperature	1.0
Training steps	299
Save interval	64 steps
Objective	On-policy distillation
Teacher supervision	Token log-probabilities
OPD coefficient	1.0
Advantage estimator	GRPO
KL loss	Disabled
Entropy bonus	Disabled
Global batch size	64
Learning rate	10^{-6}
Schedule	Constant
Optimizer	Adam
Weight decay	0.1
Adam betas	(0.9, 0.98)

TABLE 4 Default OPD setting used in the main experiments.

The bf16-aware sparsity is

$$\text{sp}_{\text{bf16}}(W_0, W_+) = 1 - \frac{1}{n} \sum_{i,j} \mathbb{1}[W_{+,ij} \not\approx_{\eta} W_{0,ij}],$$

where n is the number of entries in the analyzed matrix. Overall sparsity is parameter-count weighted across analyzed weights.

Principal-angle rotation. For each matrix, let

$$W_0 = U_0 \Sigma_0 V_0^{\top}, W_+ = U_+ \Sigma_+ V_+^{\top}.$$

Let $U_{0,k}$, $V_{0,k}$ and $U_{+,k}$, $V_{+,k}$ denote the top- k left and right singular subspaces. We compute principal angles by

$$\cos \theta_i^U = \sigma_i(U_{0,k}^{\top} U_{+,k}), \cos \theta_i^V = \sigma_i(V_{0,k}^{\top} V_{+,k}).$$

We use $k = 512$ by default and report summary statistics of the resulting angle vectors. These diagnostics use CPU float64 SVD, since GPU SVD can produce unstable angles for near-identical matrices.

Item	RLVR default setting
Initialization	SFT anchor iter_0005375
Student	Qwen3-8B
Teacher	None
Training data	dapo-math-17k
Rollout policy	Student-generated, on-policy
Rollout prompts	1024 per step
Samples per prompt	8
Rollout temperature	1.0
Training steps	1023
Objective	Verifier-reward GRPO
Reward	Binary DeepScaler math accuracy
Advantage estimator	GRPO
KL loss	Disabled
Entropy bonus	Disabled
PPO clipping	$\epsilon = 0.2, \epsilon_{\text{high}} = 0.28$
Dynamic filter	Skip batches with zero reward variance
Global batch size	512
Learning rate	10^{-6}
Schedule	Constant
Optimizer	Adam
Weight decay	0.1
Adam betas	(0.9, 0.98)
Adam epsilon	10^{-15}

TABLE 5 Default RLVR setting used in the main experiments.

Spectral drift. Let $\sigma(W)$ denote the singular-value vector of W in descending order. We measure normalized spectral shift as

$$\text{NSS}(W_0, W_+) = \frac{\|\sigma(W_+) - \sigma(W_0)\|_2}{\|\sigma(W_0)\|_2}.$$

NSS is computed per matrix and then averaged without parameter-count weighting.

Update-mask overlap. The visible update mask is

$$M_{\text{upd}} = \{(i, j) : W_{+,ij} \not\approx_{\eta} W_{0,ij}\}.$$

We compare it with two masks derived from W_0 . The principal mask M_{princ} contains the top- α fraction of entries by magnitude in the rank- r SVD reconstruction of W_0 , and the low-magnitude mask M_{low} contains the bottom- α fraction of entries by $|W_0|$. For $M_{\star} \in \{M_{\text{princ}}, M_{\text{low}}\}$, we report

$$\text{Overlap}(M_{\star}, M_{\text{upd}}) = \frac{|M_{\star} \cap M_{\text{upd}}|}{|M_{\text{upd}}|}.$$

Variant	Student	Teacher	Changed factor
Baseline	Qwen3-8B	Qwen3-32B	Reference
MoE teacher	Qwen3-8B	Qwen3-30B-A3B	Teacher architecture
4B→8B	Qwen3-4B	Qwen3-8B	Teacher scale
4B→14B	Qwen3-4B	Qwen3-14B	Teacher scale
4B→32B	Qwen3-4B	Qwen3-32B	Teacher scale
14B→32B	Qwen3-14B	Qwen3-32B	Student scale
Code domain	Qwen3-8B	Qwen3-32B	Data domain
Multi-seed	Qwen3-8B	Qwen3-32B	Random seed

TABLE 6 OPD variant runs used for robustness checks in Table 1.

Variant	Additional setting
Code domain	Uses DeepCoder data and LiveCodeBench v5 evaluation
MoE teacher	Uses Qwen3-30B-A3B as the teacher model
Multi-seed	Uses the same Qwen3-8B/Qwen3-32B math setting with different random seeds

TABLE 7 Additional notes for OPD variant runs.

We use rank $r = 64$ and $\alpha = 0.5$ unless otherwise specified. The random baseline is α , since each reference mask contains an α fraction of entries.

Matrix coverage and averaging. For bf16-aware sparsity, we compute overall sparsity over all analyzed weight parameters and report both overall and per-type summaries. For spectral geometry and update-mask overlap, we analyze standard weight matrices across layers; QKV projections are split into Q, K, and V when needed. Metrics are first computed per matrix and then averaged across matrices. This avoids domination by a small number of large matrices while preserving layer-wise and module-wise variation.

D Objectives and Three-Gate Extension

Original Three-Gate account. The Three-Gate account of RLVR [24] explains visible update sparsity as the outcome of a constrained, geometry-steered, and precision-filtered optimization process.

Gate I: KL anchor. RLVR updates are locally constrained in policy space. In a KL-regularized or trust-region view, the post-update policy remains close to the current or reference policy:

$$D_{\text{KL}}(\pi_{\theta^+} \parallel \pi_{\theta}) \leq K. \quad (15)$$

This policy-space leash prevents each step from freely rewriting the pretrained model.

Gate II: model geometry. The KL anchor limits the size of the update, but not its location.

The second gate attributes this location to pretrained model geometry. Given a bounded step, the structured loss landscape steers updates away from high-curvature principal directions and toward lower-curvature, spectrum-preserving regions. This explains why RLVR updates are off-principal rather than merely small.

Gate III: precision realization. The final gate determines which coordinates become visible in stored weights. Under bf16 precision, sub-threshold micro-updates are not realized as changed parameters. Thus the underlying routing bias appears as high bf16-aware update sparsity: preferred regions receive visible updates, while changes elsewhere remain hidden.

Training objectives. We write the three post-training paradigms in a common notation. For an input x and sequence $y = (y_1, \dots, y_T)$, SFT minimizes cross-entropy on offline demonstrations:

$$\mathcal{L}_{\text{SFT}} = - \sum_{t=1}^T \log p_{\theta}(y_t^{\star} \mid x, y_{<t}^{\star}). \quad (16)$$

RLVR uses on-policy samples with a scalar sequence-level advantage:

$$\mathcal{L}_{\text{RLVR}} = -A(y) \sum_{t=1}^T \log p_{\theta}(y_t \mid x, y_{<t}) + \beta \mathcal{R}_{\text{KL}}, \quad (17)$$

where

$$\mathcal{R}_{\text{KL}} = D_{\text{KL}}(p_{\theta}(\cdot \mid x) \parallel p_{\text{ref}}(\cdot \mid x)). \quad (18)$$

Here $A(y)$ denotes a normalized sequence-level advantage and p_{ref} is the reference policy. Clip-only RL variants can be viewed as replacing the explicit KL penalty with a local trust-region effect.

OPD optimizes student-generated rollouts against a teacher distribution:

$$\mathcal{L}_{\text{OPD}} = \sum_{t=1}^T D_{\text{KL}}(p_{\theta}^t \parallel q_T^t), \quad (19)$$

where

$$p_{\theta}^t = p_{\theta}(\cdot \mid x, y_{<t}), q_T^t = q_T(\cdot \mid x, y_{<t}).$$

If a forward-KL implementation is used, the KL direction is replaced accordingly; the signal-granularity argument below is unchanged.

Unified score-weighted form. Let

$$\phi_t = \nabla_{\theta} \log p_{\theta}(y_t \mid x, y_{<t}) \quad (20)$$

denote the token score. A broad class of stochastic post-training updates can be abstracted as

$$g = \sum_{t=1}^T a_t \phi_t. \quad (21)$$

The distinction lies in the coefficient a_t . In RLVR, $a_t = A(y)$ is shared across the sequence. In OPD, a_t varies across tokens, induced by teacher-student disagreement. Thus OPD keeps the on-policy structure of RLVR but replaces a scalar credit signal with dense token-level supervision.

OPD-specific relaxation. The relaxation appears at the level of update covariance. Let $C = \mathbb{E}[gg^\top]$. For RLVR,

$$C_{\text{RLVR}} = \mathbb{E}[A(y)^2 ss^\top], s = \sum_t \phi_t. \quad (22)$$

For OPD,

$$C_{\text{OPD}} = \mathbb{E}\left[\sum_{t,t'} a_t a_{t'} \phi_t \phi_{t'}^\top\right]. \quad (23)$$

RLVR couples all token scores through one sequence-level scalar. OPD uses heterogeneous token coefficients, expanding the accessible directional support within the same geometry-steered update family. This is the sense in which OPD is *relaxed*: it remains anchored and geometry-constrained, but less selective than RLVR.

Local anchor and weight bound. We now make the anchoring step explicit. For a weight block W , consider a local quadratic constraint

$$\frac{1}{2} \langle \text{vec}(\Delta W), S_W \text{vec}(\Delta W) \rangle \leq \delta_W, S_W \geq \mu_W I. \quad (24)$$

This implies

$$\|\Delta W\|_F \leq \sqrt{\frac{2\delta_W}{\mu_W}}. \quad (25)$$

The bound captures the anchoring effect. OPD differs from RLVR not by removing the anchor, but by using denser token-level information inside the anchored region.

Spectral stability under bounded updates. Let $W_+ = W_0 + \Delta W$ and $\gamma_k = \sigma_k(W_0) - \sigma_{k+1}(W_0)$ be the singular-value gap. By Wedin-type perturbation bounds,

$$\begin{aligned} & \max\{\|\sin \Theta(U_k(W_0), U_k(W_+))\|_2, \\ & \|\sin \Theta(V_k(W_0), V_k(W_+))\|_2\} \leq \frac{\|\Delta W\|_2}{\gamma_k}. \end{aligned} \quad (26)$$

Singular values satisfy

$$|\sigma_i(W_+) - \sigma_i(W_0)| \leq \|\Delta W\|_2, \quad (27)$$

and

$$\|\sigma(W_+) - \sigma(W_0)\|_2 \leq \|\Delta W\|_F. \quad (28)$$

Therefore, updates with small operator and Frobenius norm tend to preserve the pretrained spectral structure. OPD remains in this bounded regime, but its dense token-level signal permits larger subspace rotation and spectral drift than RLVR.

Precision realization. bf16 precision determines which coordinates become visible in stored weights. A coordinate is realized as changed only if

$$|\Delta W_{ij}| \gtrsim \frac{1}{2} \text{ULP}_{\text{bf16}}(W_{0,ij}). \quad (29)$$

The corresponding visible update mask is

$$M_{ij} = 1 \left[|\Delta W_{ij}| \gtrsim \frac{1}{2} \text{ULP}_{\text{bf16}}(W_{0,ij}) \right]. \quad (30)$$

Since OPD updates are larger than RLVR updates but remain geometry-constrained, more coordinates cross the realization threshold, yielding intermediate bf16-aware sparsity.

E Trajectory Metric Definitions

For trajectory-level analysis, we study the cumulative update $\Delta W_t = W_t - W_0$ at each checkpoint. Unless otherwise specified, metrics are computed for each analyzed weight matrix and then averaged across matrices.

Stable rank. Stable rank measures the effective number of dominant singular directions that carry the update energy:

$$\text{srank}(\Delta W_t) = \frac{\|\Delta W_t\|_F^2}{\|\Delta W_t\|_{\text{op}}^2}. \quad (31)$$

Unlike algebraic rank, stable rank is insensitive to arbitrarily small singular values and therefore provides a scale-aware measure of effective update dimensionality.

Frobenius norm. We use the Frobenius norm to measure cumulative update magnitude:

$$\|\Delta W_t\|_F = \left(\sum_{i,j} (\Delta W_{t,ij})^2 \right)^{1/2}. \quad (32)$$

This diagnostic rules out the possibility that low stable rank is merely caused by negligible parameter movement.

Hill tail estimate. To inspect the spectral shape of ΔW_t , we use the Hill tail estimator [6]. Let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k$ denote the selected top singular values. We estimate the tail exponent as

$$\hat{\alpha}_{\text{Hill}} = \left[\frac{1}{k-1} \sum_{i=1}^{k-1} \log \frac{\sigma_i}{\sigma_k} \right]^{-1}. \quad (33)$$

We use this only as an auxiliary spectral-shape diagnostic, complementary to stable rank.

F Additional Evaluation of Rank-Constrained Training

To test whether the functional-sufficiency result in Section 4.2 is specific to AIME 2024, we evaluate the same rank-16 projected runs on additional reasoning benchmarks, as shown in Figure 8. The projection subspace, projection start, and matched training windows are the same as in the main experiment.

The additional benchmarks show the same qualitative pattern as Figure 6. OPD remains broadly robust under the early rank-16 subspace constraint, whereas SFT is substantially more sensitive to the same bottleneck. This supports the conclusion that the early low-dimensional update channel is functionally sufficient for OPD beyond the primary AIME 2024 evaluation.

G Control Experiment Details

Item	Setting
Student model	Qwen3-8B
Teacher model	Qwen3-32B
Student initialization	Qwen3-8B SFT anchor, iter_0005375
Training data	dapo-math-17k
Training length	300 steps
Checkpoint steps	63, 127, 191, 255, 299
Baseline rollout	Student-generated, 8 samples per prompt
Analyzed matrices	36 layers $\times$ 4 attention weights
Primary diagnostic	Stable rank
Auxiliary diagnostics	Frobenius norm, Hill tail estimate

TABLE 8 Shared setup for OPD control experiments.

G.1 Shared OPD Control Setup

All control experiments in Section 5 are based on the same OPD baseline configuration. Unless otherwise specified, each intervention changes only the stated factor and keeps the remaining OPD training pipeline fixed. Full training hyperparameters are provided in Appendix C.

For all spectral diagnostics, we analyze the same matrix set as in the main trajectory analysis: 144 matrices, corresponding to 36 layers and four attention weight types per layer. Metrics are computed per matrix and then averaged across matrices.

G.2 Perturbation Protocols

Token sparsification. This intervention perturbs the density of token-level teacher supervision while keeping the rollout policy and OPD objective fixed. For each sampled response token a_t , the implementation computes the sampled-token student-teacher log-probability gap

$$r_t = \log \pi_\theta(a_t | s_t) - \log \pi_T(a_t | s_t),$$

where π_θ is the student policy and π_T is the teacher policy. This is a sampled-token log-probability gap, not a full-vocabulary KL divergence.

For token sparsification, we introduce a binary token mask $m_t \in \{0, 1\}$. In top-KL retention, tokens are ranked by r_t and the largest ρ fraction is retained; this selects tokens where the student is most over-confident relative to the teacher. In random retention, $m_t = 1$ for a uniformly sampled ρ fraction of response tokens. We use $\rho \in \{0.25, 0.50\}$.

The retained-token signal is compensated by

$$\tilde{r}_t = \frac{T}{\sum_{\tau=1}^T m_\tau} m_t r_t.$$

The additive OPD update then uses this gap as a correction term, subtracting it from the advantage. Thus, the top-KL variant selects the tokens with the largest student–teacher deviation, while the training signal penalizes this deviation. Masking is applied independently for each sample at the response-token level. All other settings, including teacher model, on-policy rollout generation, optimizer, batch size, learning rate, and training data, are kept identical to the OPD baseline. The sparsification runs are evaluated at checkpoints $s = \{63, 127, 191, 255\}$.

Off-policy rollouts. This intervention perturbs the rollout policy while keeping the token-level OPD objective fixed. In the baseline, rollouts are generated by the current student policy:

$$y \sim \pi_\theta(\cdot \mid x).$$

In the off-policy variant, rollouts are generated by the teacher:

$$y \sim \pi_T(\cdot \mid x),$$

and the student then computes log-probabilities on these teacher-generated tokens during the training forward pass. The training objective remains pure OPD KL distillation; no reward mixing is introduced. Thus, this intervention changes the sampling distribution but keeps the teacher-token gradient source fixed.

The off-policy runs use Qwen3-32B teacher rollouts served through an sglang router with two teacher replicas. The importance-ratio clipping threshold is set to $\epsilon = 1.0$, making clipping effectively inactive for this comparison. The off-policy checkpoints are evaluated at $s = \{63, 127, 191, 255, 299\}$.

Objective interpolation. This intervention perturbs the objective composition while keeping the student rollout policy fixed. We implement this at the advantage level by mixing the OPD teacher-correction signal with the RLVR reward signal:

$$A_i^{(\alpha)} = \alpha A_{i,\text{OPD}} + (1 - \alpha) A_{i,\text{RLVR}}, \quad (34)$$

where $A_{i,\text{OPD}} = -r_i$ and $r_i = \log \pi_\theta(a_i \mid s_i) - \log \pi_T(a_i \mid s_i)$ is the sampled-token student–teacher log-probability gap. Thus, the OPD term encourages correction toward the teacher. $A_{i,\text{RLVR}}$ is the GRPO advantage from the math accuracy reward, broadcast to response tokens. The resulting policy-gradient direction is

$$g_\alpha = \mathbb{E}_i \left[A_i^{(\alpha)} \nabla_\theta \log \pi_\theta(a_i \mid s_i) \right]. \quad (35)$$

We evaluate $\alpha \in \{0.75, 0.50, 0.25, 0.05, 0.01\}$, with $\alpha = 1$ corresponding to the pure OPD baseline and $\alpha = 0$ corresponding to the RLVR endpoint. All alpha-mixing runs use raw signal mixing without per-batch standard deviation normalization. The rollout policy remains on-policy, and the optimizer, batch size, learning rate, training data, and evaluation setup are identical to the baseline.

Controlled variables. Table 9 summarizes which factor is changed by each intervention.

Dimension	Baseline	Token sparse	Off-policy	Alpha mix
Rollout policy	Student	Student	Teacher	Student
Token coverage	100%	25/50%	100%	100%
Objective signal	OPD	OPD	OPD	OPD/RLVR mix
Reward module	OPD	OPD	OPD	Mixed reward
Other settings	–	Same	Same	Same

TABLE 9 Controlled variables in Section 5. Each intervention changes one target factor while keeping the remaining OPD setup fixed where possible.

G.5 Auxiliary Metrics

Stable rank is the primary diagnostic in Section 5 because the question is whether the locked low-dimensional update channel is preserved. We additionally track update scale and spectral shape through Frobenius norm and Hill tail estimates (Figure 9). These auxiliary diagnostics verify that runtime perturbations preserve the OPD-like spectral trajectory up to scale shifts, whereas objective interpolation changes the update geometry more substantially.

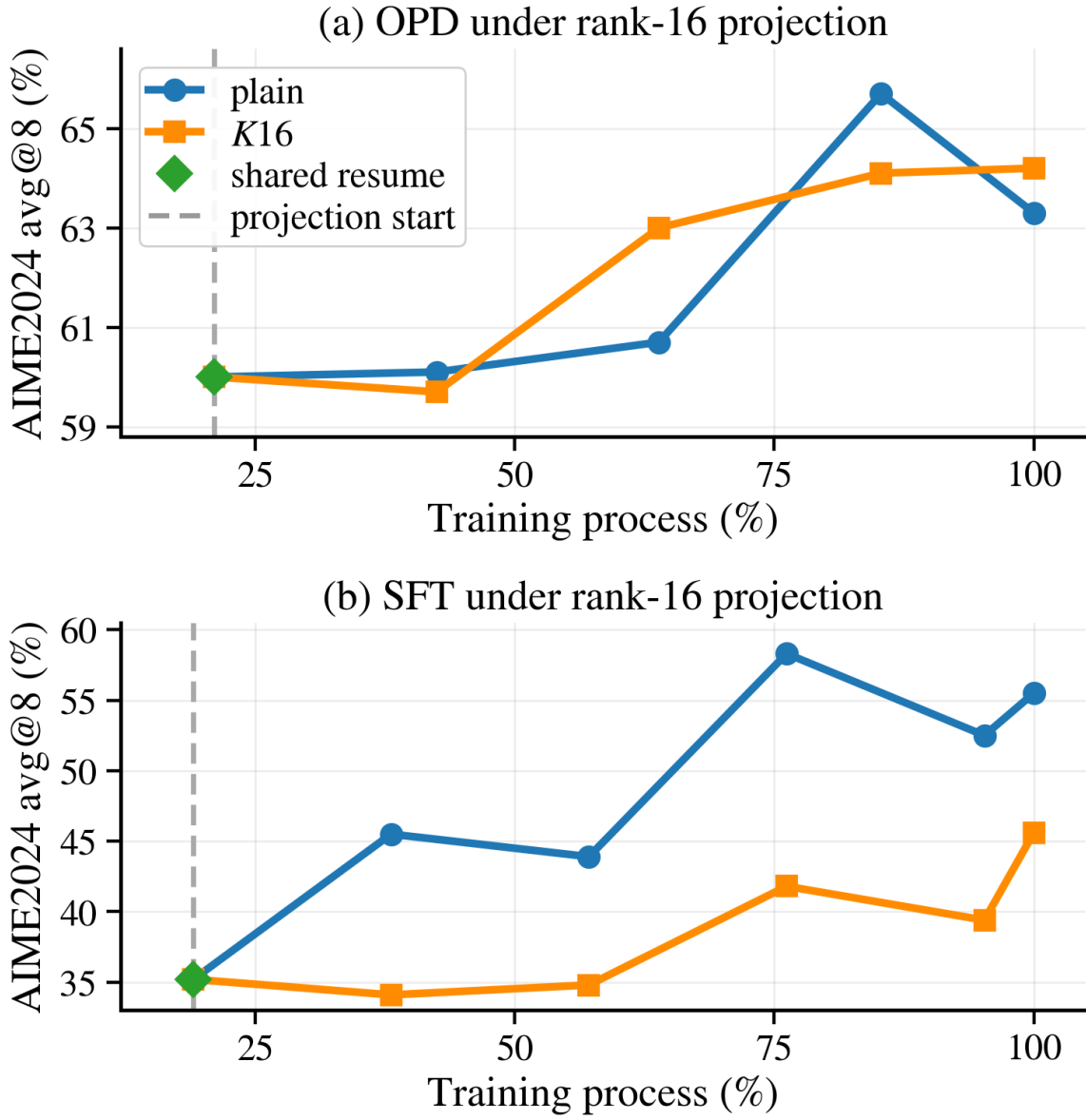


FIGURE 6 Rank-16 projected training. Rank-16 projection leaves OPD intact but degrades SFT.

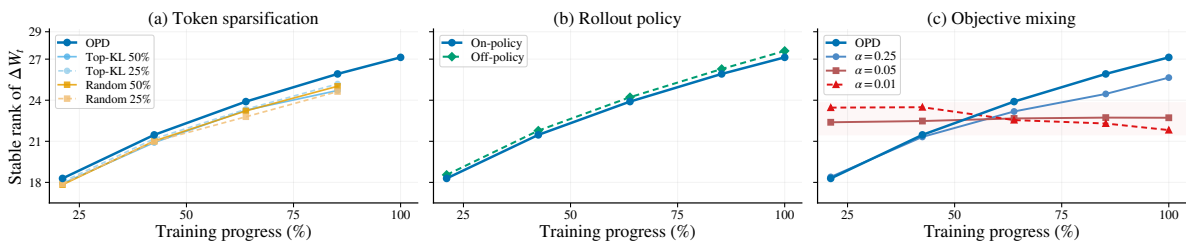


FIGURE 7 Controls on subspace locking. Runtime perturbations preserve the OPD stable-rank trajectory, whereas objective-level interpolation changes it, identifying objective composition as the sensitive control axis.

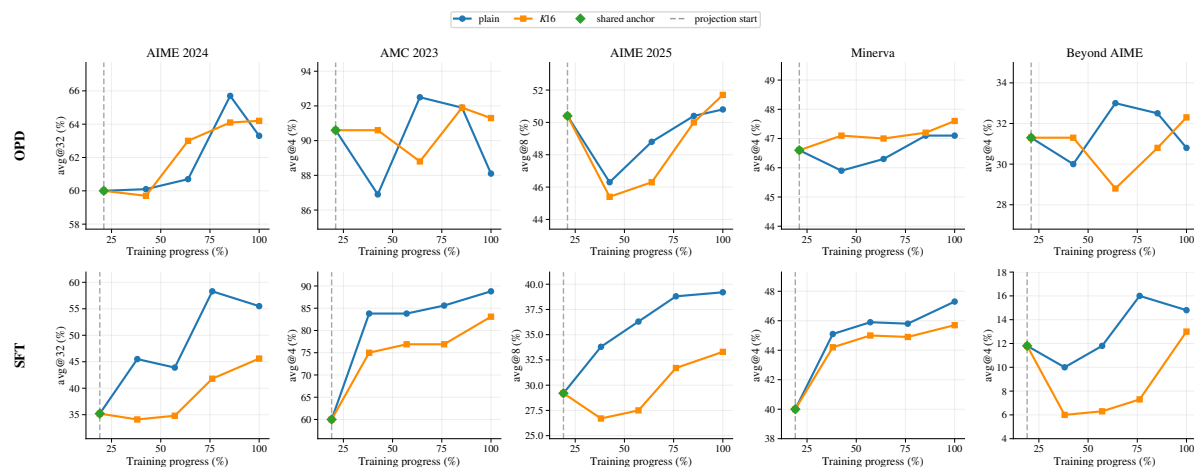


FIGURE 8 Additional evaluation of rank-constrained training. We compare unconstrained training and rank-16 projected training across five reasoning benchmarks. Across benchmarks, OPD is consistently less affected by the rank-16 bottleneck than SFT. Green diamonds denote the shared anchor checkpoint, and dashed vertical lines denote the projection start.

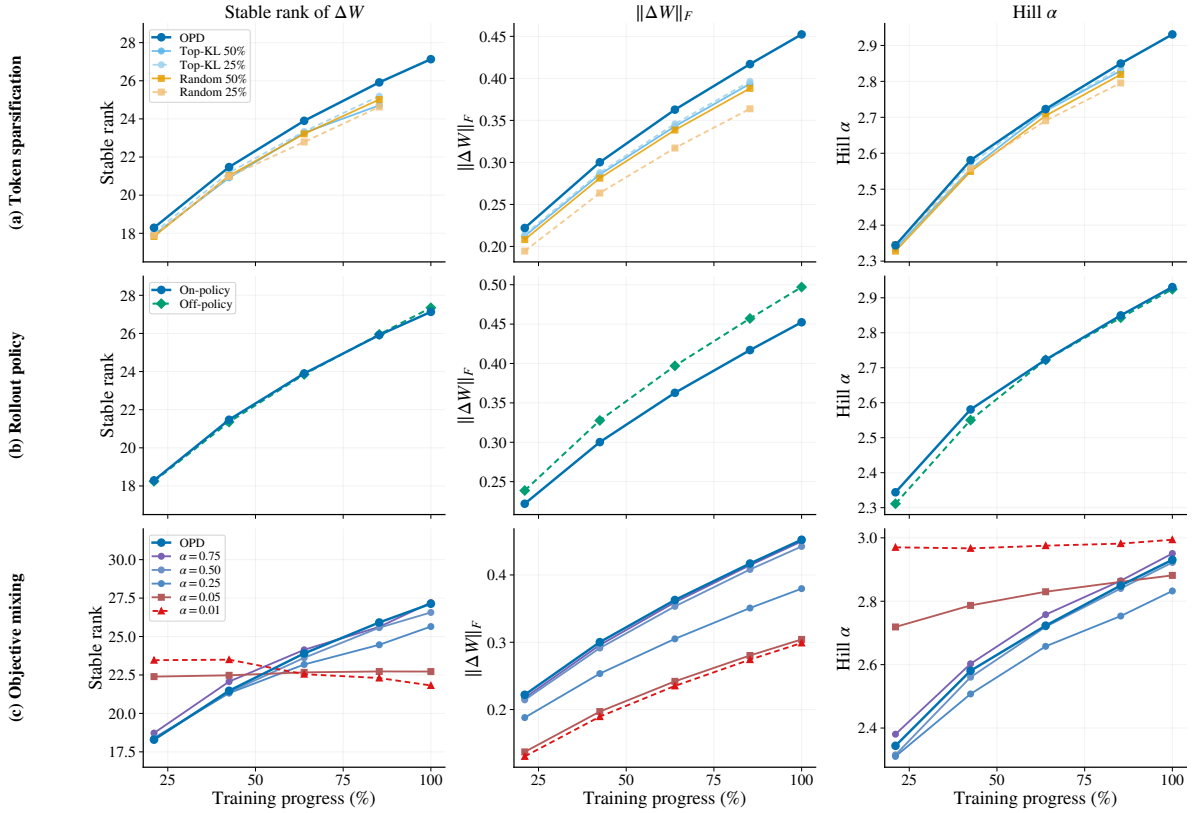


FIGURE 9 Auxiliary metrics for control experiments. We report update scale and spectral-shape diagnostics for the same perturbations analyzed in Figure 7. Runtime perturbations preserve the OPD-like spectral profile up to modest scale changes, whereas objective interpolation induces a distinct trajectory.